EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

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Subject: Low Code Application Development using Zoho Creator

Aim

To implement and explore the security, access-control, data-sharing and governance features of the EcoCycle application in Zoho Creator.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person may do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of design.

Zoho Creator provides built-in mechanisms for this. User permissions define the operations a user may perform on the components of an application. Role-based access places users in a hierarchy that influences which records they can see. Data sharing rules extend the visibility of records from one role or user to another at a defined level of access. Administrative controls allow the application to be configured and managed. For EcoCycle, a recycling officer needs to see the collection requests handled by the coordinator but does not need to change them.

Application Used

The practical was carried out in Zoho Creator, where the security and governance features of the application editor were explored for EcoCycle. The settings studied (User Permissions, Roles and Data Sharing) are available in every Zoho Creator application. At the time of the project report EcoCycle had a single owner account with no roles or permission profiles defined, so the roles and the sharing rule below are planned and are described as such.

Security and Governance Architecture

Figure 1 is a conceptual diagram of the security and governance structure of EcoCycle. It is not a screenshot of the Zoho Creator interface.

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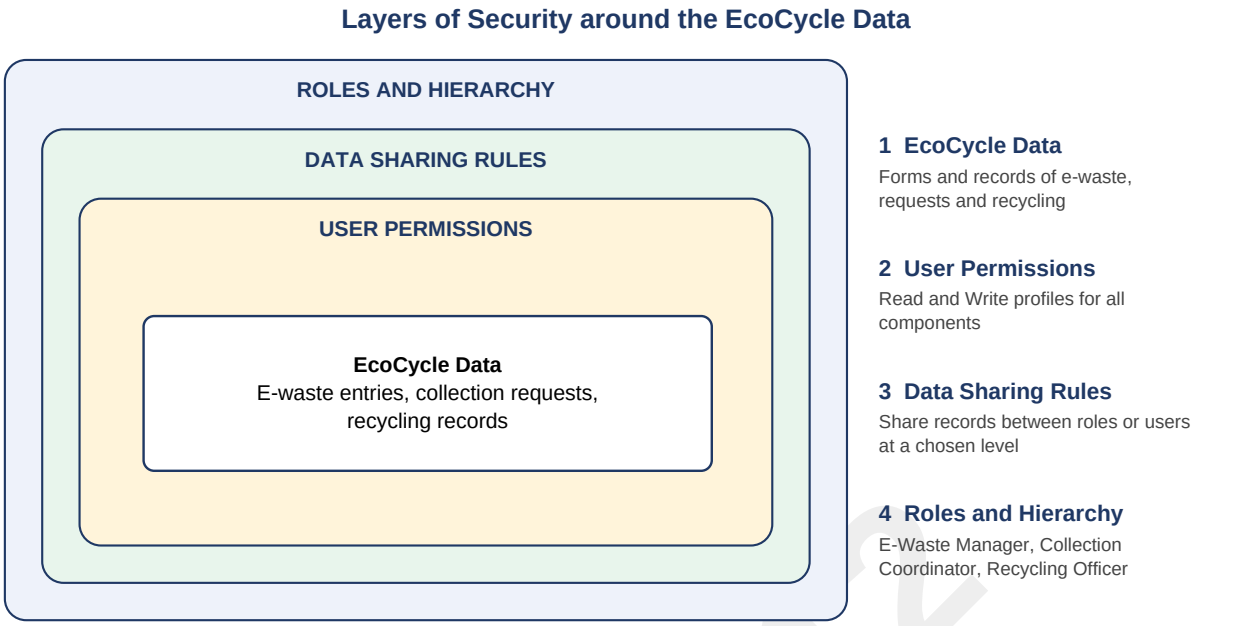

Figure 1: Security & Governance Layers (Conceptual Diagram)

The diagram shows the EcoCycle data at the core, protected by successive layers. User permissions (Read and Write) define what a user may do. Data sharing rules extend the visibility of records between roles or users at a chosen access level. The role hierarchy is the outermost layer and influences which records each role can see. The outcomes are controlled access, role-based visibility, safe collaboration and efficient administration.

User Permissions

User Permissions is opened from Settings in the application editor. It has the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records. The Permissions tab lists the permission sets of the application and offers an Add Permission option. A new application normally lists two predefined profiles:

Name	Description (as displayed)
Read	This profile will have read permission for all components
Write	This profile will have write permission for all components

[Paste your own screenshot here]

Settings → User Permissions → Permissions tab of EcoCycle

Figure 2: User Permissions (Permissions tab) in the Zoho Creator application editor

Read permission lets a user view the components and records without changing them, whereas Write permission lets the user add or modify records. A further profile, for example one for the collection coordinators of EcoCycle, can be created with Add Permission.

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. A new account shows the default top-level role CEO, and the options Expand All, Collapse All, Manage Role Hierarchy and Add Role are used to build the hierarchy.

[Paste your own screenshot here]

Settings → User Permissions → Roles tab of EcoCycle

Figure 3: Roles and Role Hierarchy (Roles tab) in the Zoho Creator application editor

For EcoCycle the following role hierarchy is planned. The roles are added with Add Role and arranged with Manage Role Hierarchy; they have not yet been created in the account.

Planned Role	Reports To	Position in Hierarchy
E-Waste Manager	–	Top level of the hierarchy
Collection Coordinator	E-Waste Manager	Second level
Recycling Officer	Collection Coordinator	Third level (lowest role)

Arranging roles this way gives EcoCycle a clear reporting structure. The position of a role in the hierarchy influences which records its members can see, and further access can be given to a specific role through data sharing rules.

Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user the records are shared from, the role or user they are shared to, and the level of access. Records then become visible to additional roles beyond the default visibility of the hierarchy. The rule planned for EcoCycle is:

Parameter	Planned Value
Rule Name	Collection_Request_Share
Form	Collection Request
Records Shared From	Roles – Collection Coordinator
Records Shared To	Roles – Recycling Officer
Level of Access	Read Only

[Paste your own screenshot here]

Settings → User Permissions → Data Sharing → New Share Rule panel

Figure 4: New Share Rule panel (Data Sharing tab) in the Zoho Creator application editor

The New Share Rule panel has the fields Name, Form, Records shared from, Records shared To and Level of access (Read Only by default). The “Records shared from” field offers Roles, Roles and Subordinates, and Users. Roles shares the records of one selected role, Roles and Subordinates adds the roles beneath it, and Users shares the records of individual users. With the planned rule, members of the Recycling Officer role could view the shared Collection Request records but could not modify them. The rule is planned until the roles exist in the account.

Security Features and Governance Controls

Figure 5 classifies the security and governance controls of Zoho Creator as they apply to EcoCycle. It is a conceptual diagram, not a screenshot, and it is not proof that every listed feature was configured.

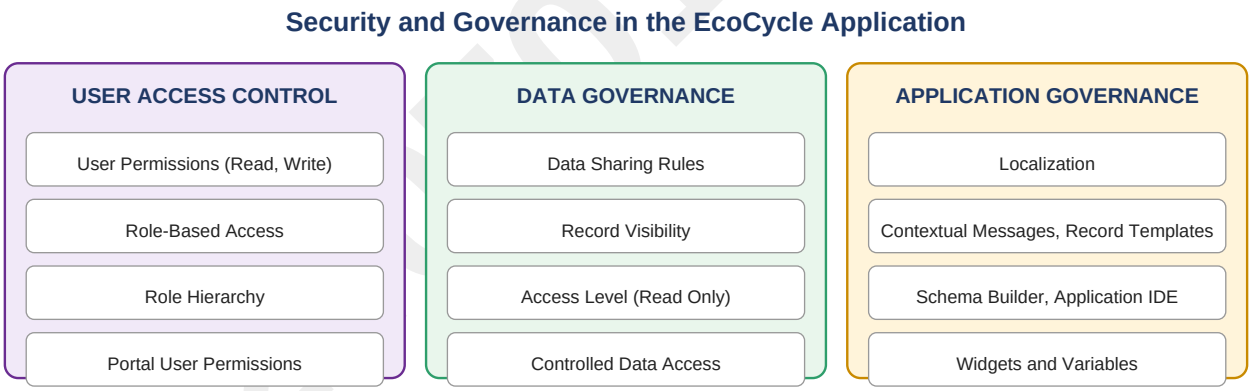

Figure 5: Classification of Security and Governance Controls (Conceptual Diagram)

Control	Explanation
User access control	User permissions (Read and Write), role-based access and portal user permissions decide who can use the application and what they can do.
Data governance	Data sharing, access level, record visibility and the role hierarchy govern how application data is accessed.
Application governance	Localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	A reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another at a chosen access level.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

Security controls already present in EcoCycle

- The **EcoCycle Data Assistant** chat agent has read-only access, so it can answer questions about high-priority e-waste, pending requests and recycling status without changing any record.
- The **EcoCycle Summary API** is a Custom API with a GET endpoint secured with OAuth2, so a caller must present a valid access token.
- A **past-date check** on the Collection Request form rejects a preferred collection date earlier than today, and mandatory fields carry a red asterisk.
- Access currently relies on the default access of the owner account; roles and permission profiles are listed as future work.

In this practical, user permissions and roles were examined and the New Share Rule panel was opened. The other items in the diagram, such as portal user permissions, localization, schema builder and widgets, are available in the Settings menu but were not configured in this experiment.

Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and controls the access of external portal users. It was not configured in this experiment, and no portal permissions were set for EcoCycle.

Role-Based Access and Data Sharing Flow

Figure 6 combines the planned role hierarchy of EcoCycle with the planned data sharing rule. On the left the roles are shown in reporting order. On the right the rule Collection_Request_Share is traced from the Collection Request form through the role it is shared from, to the role it is shared to and the level of access.

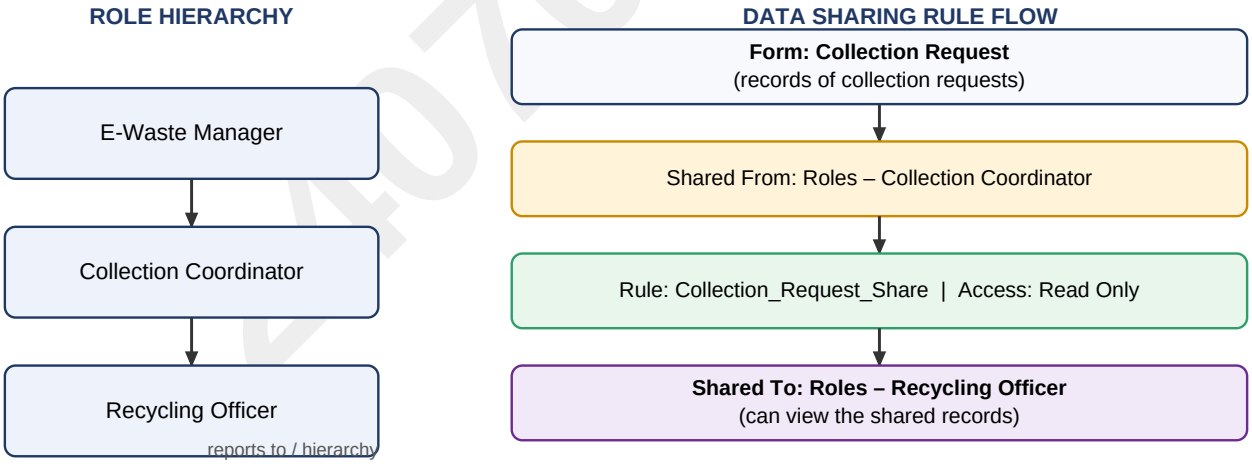

Figure 6: Role Hierarchy and Data Sharing Flow (Conceptual Diagram)

Result: the Recycling Officer role would receive Read Only access to the Collection Request records shared by the Collection Coordinator role.

- The roles planned for EcoCycle form a hierarchy: E-Waste Manager, Collection Coordinator and Recycling Officer.
- The data sharing rule Collection_Request_Share is planned for the Collection Request form.
- Records would be shared from the Collection Coordinator role to the Recycling Officer role.
- The level of access is Read Only, so the Recycling Officer role could view the shared records but not modify them.

Governance Observations

- Predefined Read and Write permission profiles are available in the application.
- The Roles tab shows the default role; the EcoCycle roles are planned and are added with Add Role.
- The Data Sharing tab lists no rule until one is added; the New Share Rule panel was opened to study its fields.
- Portal User Permissions was not configured in this experiment.
- The practical showed how access can be limited to what a role needs by choosing a suitable level of access.

Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment the Read and Write permission profiles were examined, the role tree of the application editor was viewed, and the New Share Rule panel of the Data Sharing tab was opened. A role hierarchy of E-Waste Manager, Collection Coordinator and Recycling Officer, and a data sharing rule named Collection_Request_Share, sharing Collection Request records from the Collection Coordinator role to the Recycling Officer role with Read Only access, were planned for EcoCycle.

The experiment shows how permissions, roles and data sharing work together so that each role sees the records it needs and no more. Choosing a suitable level of access, such as Read Only, allows records to be shared for collaboration without allowing them to be changed. Together with the read-only chat agent and the OAuth2-secured Custom API already present in EcoCycle, controlling who can reach application data, and at what level, remains an essential part of building and managing business applications.

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